Preface

In this research, I conducted a thorough study and modelling of movie success prediction and content-based recommendations. The fundamental goals of this project are threefold: to analyse and comprehend the properties of movie data, to construct models that forecast whether a movie will be a hit or a flop, and to create a content-based recommendation system for proposing movies.

To achieve these goals, I gathered and preprocessed a large dataset of movie parameters such as genre, cast, director, budget, revenue, and more. This dataset was used as the foundation for exploratory data analysis (EDA), in which I sought to comprehend the data's patterns and trends. Through visualisation and statistical analysis, I found significant factors that might impact a film's success.

For the predictive modelling assignment, I used a number of machine learning techniques to categorise films as hits or flops. These algorithms are Logistic Regression, Decision Trees, Random Forests, XGBoost, and Neural Networks. Each model was trained and verified using relevant metrics to guarantee reliable performance. Logistic Regression supplied a baseline model with interpretable coefficients, but Decision Trees and Random Forests revealed feature significance and relationships. XGBoost's gradient boosting method improved predicted accuracy, whereas Neural Networks caught intricate non-linear correlations in the data.

In addition to predictive modelling, I created a content-based recommendation system that suggests films based on content similarities. Using techniques like TF-IDF vectorization and cosine similarity, I created a recommendation engine that finds and recommends comparable movies based on the movie's summary and other textual elements. This algorithm improves the user encounter by recommending films based on content and individual interests. These algorithms and approaches collaborate to provide a complete and successful framework for assessing and forecasting movie efficiency, as well as providing customers with tailored movie recommendations.

Overall, my project displays my ability to use sophisticated machine learning and natural language processing techniques to solve real-world entertainment business difficulties. The models and techniques described herein have the ability to offer stakeholders with vital insights while also increasing user engagement through excellent movie recommendations.